TIC 290436243

Eclipsing binary candidate dossier - Star Vetter

THIN MARGIN

```
BLS EPHEMERIS
  Period        1.079330 d
  Epoch (t0)    1625.8121 BTJD
  Duration      0.1200 d
  Depth         3.352 %  (33,522 ppm)
  SNR           113.0

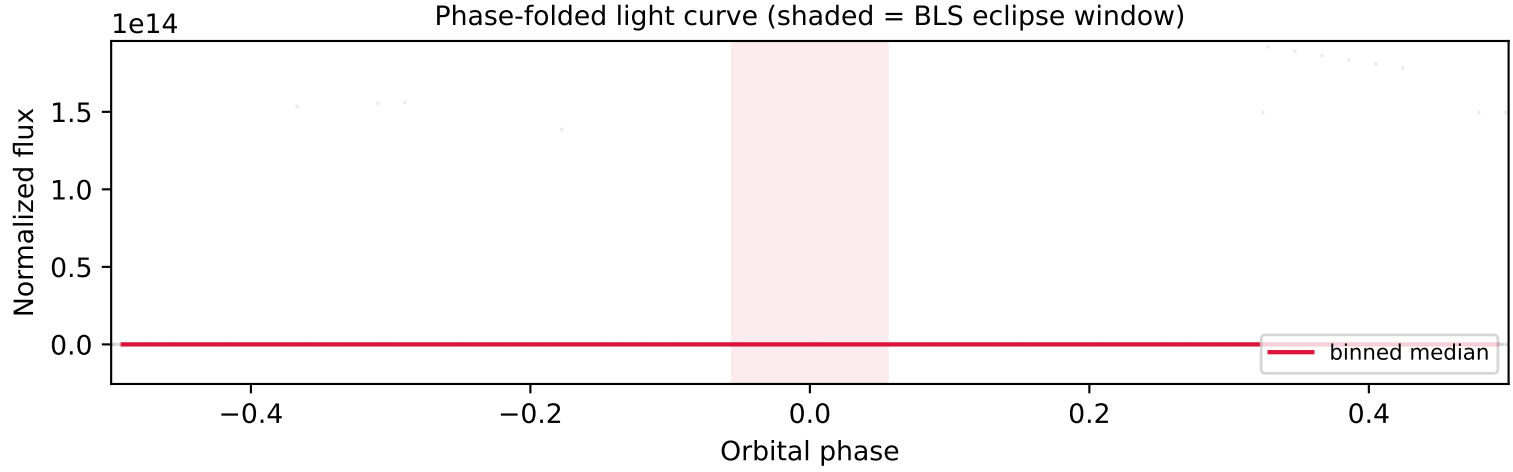
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed      23  (gate >= 3)
  Robust depth ratio     0.95  (gate >= 0.50)
  Robust depth sigma     15.0  (gate >= 7.0)
  Odd/even z-score       0.13  (consistent)
  Secondary eclipse      not detected
```

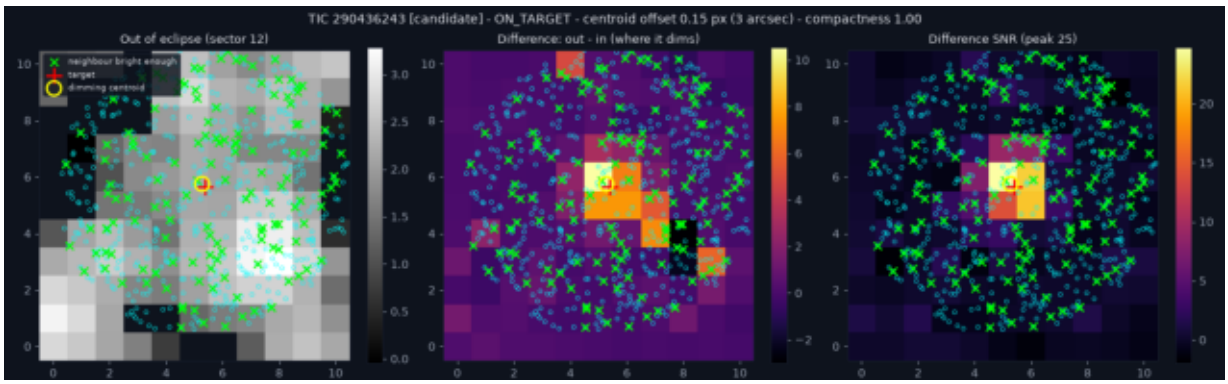
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CATALOG CROSS-MATCH (novelty)

  VSX                                 no match
  ASAS-SN                             no match
  Gaia eclipsing-binary table         no match
  Gaia variable classification        no match
  TESS EB catalog (Prsa+2022)         no match
  ExoFOP TOI list                     no match

  Verdict: NOVEL
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PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 12    light-curve dip in this sector: 20.8 sigma    t0 refined by -0.0025 phase
  Difference-image peak SNR   24.78   (gate >= 5.0)      Compactness   1.00   (gate >= 0.50)
  Dimming centroid offset     0.15 px (3 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 290436250
    0.23 px away, dTmag +3.43 - farther from the dimming than the target
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Caveat (THIN MARGIN): The ON_TARGET call rests on a neighbour-competition margin of just 0.071 px (dimming centroid 0.154 px from target vs 0.225 px from the nearest bright-enough neighbour) - inside diag_pixel.py's own stated centroid systematic of 0.10-0.3 px. TESS's resolution cannot confidently separate this star from that neighbour.